

Youssef Anbar

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OBJECTIVE

Seeking to contribute to integrated hardware and power systems at scale, building on PCB design, embedded programming, and Python/C++ software.

EDUCATION

The University of Texas at Austin

Bachelor of Science in Electrical and Computer Engineering, Honors

Coursework: Circuit Theory, Embedded Systems, Software Design, Intro to Computing, Intro to Electrical Engineering

GPA: 4.0/4.0 | Dean's List (Fall 2025, Spring 2026)

May 2028

Austin, TX

The Village School (High School), Valedictorian (4.8/4.0 GPA), Rank 1/196

High School Diploma / International Baccalaureate (IB) Diploma Programme

- IB Score: 44/45 (Top 0.5% of Students Globally)

May 2025

Houston, TX

WORK EXPERIENCE

GE Vernova, ITR / Design Engineering Intern, Houston, TX

May 2026 - Aug 2026

- Built an AI cross-document verification system spanning 7 document types and 6 automated consistency checks; projected 93% reduction in review labor and cycle time from 16 to 4 weeks across 5 departments, easing backlog through 2029
- Completed 40+ design review checks and documented 20+ technical anomalies across F&IDs, one-line diagrams, motor schematics, and customer wiring and load adjustments, aligning drawings to IEEE 315-1975
- Identified 5+ anomalies in Frontier engine data by correlating min-gate function software trends against EPlan modeling, root-causing each using a linear regression model
- Automated competitive benchmarking through a Velocity Suite data collector and a datacenter market intelligence scraper, replacing manual tracking with live fuel-cost and operational data on GE turbines versus competitors
- Authored a research paper on clutched versus clutchless synchronous condensers for grid stability applications

BMW, Electrical Engineer Service Intern, Houston, TX

Jul. 2024

- Aided in diagnostic tests on Digital Motor Electronics (DME) using OBD-II scanners and BMW ISTA software to troubleshoot 4+ control unit failures across 25+ BMW vehicle models
- Worked alongside senior technicians on in-circuit testing (ICT) and ECU programming, building familiarity with control module software and fault-tracing workflows on live service vehicles

Baker Hughes, Electrical Engineering Intern, Houston, TX

Jun. 2023

- Managed rotary steerable gear tests measuring drill torque under simulated downhole pressure to accelerate early-phase failure analysis; observed Ansys and Jewel Suite modeling of drill performance for electronics integration

RESEARCH / PROJECTS

Texas Spacecraft Laboratory (TSL), Command & Data Handling (CDH), NASA Partner

UT Austin

- Developing CDH flight software in C++ using NASA JPL's F' (F Prime) framework; authoring Components, Ports, and Commands for telemetry and command dispatch
- Implementing rate-group scheduled tasks and FPP model definitions for autocoded interfaces, validated on Linux/ARM target hardware

Embedded Chipless RFID Research (SHM), Undergraduate Research Assistant, PhD Lab

UT Austin

- DARPA-scale research on wireless chipless RFID strain and temperature sensors embedded in LPBF-manufactured metal parts
- Designing resonant RLC sensing architectures and supporting micro-cold spray fabrication targeting $\sim 1 \times 10^{-5}$ strain and $< 1^\circ \text{C}$ resolution

tinyCore Industries (V1), Embedded Hardware Engineer (PCB / Power), Team of 8

Pre-seed \$700K

- Designed a custom micro-drone HAT PCB in Altium Designer driving 4x brushed coreless motors via MOSFET low-side switching, with 1S LiPo/LiHV power delivery and an onboard 3.3 V LDO feeding the MCU
- Replaced motor connectors with through-hole solder pads for vibration resistance; validated manufacturability via 3D PCB export and SolidWorks fit checks; platform sold to universities

Inhabit Robotics, Co-founder (V2), Dexterous Teleop Hand + Modular CAN Joint Architecture

In talks with Tess Ventures and angel investors

- Built a dexterous teleop hand on the Unitree G1 EDU (Dex3-1) via `xr_teleoperate`; mapped operator joint poses to force-controlled actuators; demoed to SF VCs
- Architecting V2: daisy-chainable CAN-bus joints using MCUs, magnetic encoders, and SPI-CAN, with auto-enumeration so the base controller detects module order to the end-effector

Longhorn Baja Electric Racing (SAE), Electronics Racing Team

UT Austin

- Integrated 7+ sensors into a Raspberry Pi Pico ECU (Hall-effect wheel speed, 6-axis IMU, brake pressure, engine/CVT temperature)
- Implemented ~ 500 Hz acquisition per sensor with microSD telemetry logging and < 5 ms sampling latency under vibration and EMI

AWARDS & HONORS

- Yconic x NVIDIA Hackathon, 2nd Place (\$4,000): MCU-driven datacenter thermal twin modeling bidirectional API flow between heat sources and heat-pump sinks
- TVG Hackathon, 2nd Place (\$3,000): Privacy-first customer-service escalation system using Google Gemini sentiment analysis to flag escalating discussions via dashboard
- Designathon, 2nd Place (\$1,000): LiDAR and GPS smart-bin product with a dashboard to optimize trash layout and routing
- UT Austin Honors Scholarship, \$36,000 | Cockrell School of Engineering

TECHNICAL SKILLS / ADDITIONAL INFORMATION

Languages: C, C++, Python (NumPy, Pandas, Matplotlib; regression, classification, neural networks), Java, JavaScript, SQL, Swift

Tools: Altium Designer, SolidWorks, GitHub, Excel, CAD, FRC Robotics | Work Auth: Permanent Resident